

A Finite-Resonance Sieve Obstruction on Prime Shells

Liang Wang

Huazhong University of Science and Technology

August 24, 2026

Abstract

Let $\mathcal{P}_Q$ be the primes in $(Q, 2Q)$ and let $E_{3,7}(Q)$ count primitive pairs $p < r$ in $\mathcal{P}_Q$ satisfying $7p + 3r = 16Q$. We compute the complete local root law for an exact two-linear-form parameterization and apply the Selberg upper-bound sieve to prove

$$E_{3,7}(Q) \ll \frac{Q \log \log(3Q)}{(\log Q)^2}, \quad \frac{E_{3,7}(Q)}{|\mathcal{P}_Q|} \rightarrow 0.$$

The same conclusion holds for every fixed finite family of primitive nondegenerate linear resonances. Combined with an earlier sharp matched-mass ceiling, this proves that the literal aligned first-resonance mechanism cannot provide any fixed positive proportional energy saving at all sufficiently large scales, even under perfect edge anti-alignment. This is a scoped arithmetic obstruction, not a prime-pair lower bound or a cancellation theorem for the underlying physical source.

1 Introduction

The motivating prime-shell collision is

$$7p + 3r = 16Q, \quad Q < p < r < 2Q, \tag{1}$$

with the primitive condition $(Q, 21) = 1$. Earlier exact work shows that these collisions form a matching and that a proportional energy saving cannot exceed the diagonal mass carried by its incident vertices. In the literal aligned row model, strict saving by δ therefore requires

$$\frac{E_{3,7}(Q)}{|\mathcal{P}_Q|} \geq \frac{\delta}{8}. \tag{2}$$

For the program endpoint $\delta = 1/400$, the toll is $1/3200$.

The purpose of this paper is to decide whether the left side of (2) can remain bounded below. The answer is no. The relevant arithmetic is an ordinary dimension-two upper-bound sieve, but its application requires keeping the moving determinant $16Q$ and its exceptional local factors. We use the classical Selberg sieve in the forms developed in [2, 3]; the Goldbach and twin-prime derivations in [1, 4] provide close two-form templates.

Our second result isolates the real scope of the obstruction: any *fixed finite* set of nondegenerate linear resonance equations has zero edge density in the prime shell. A family whose depth grows with Q , or an arithmetic concentration of actual source mass on a sparse support, is not ruled out.

2 Exact local arithmetic

Assume $(Q, 21) = 1$ and write $Q = 3t + a$, where $a \in \{1, 2\}$. Reducing (1) modulo three gives $p \equiv a \pmod{3}$. Thus every solution has the unique parameterization

$$L_1(k) = 3k + a, \quad L_2(k) = 16t + 3a - 7k. \quad (3)$$

Its determinant is

$$3(16t + 3a) + 7a = 16Q. \quad (4)$$

The shell inequalities imply $10Q/7 < L_1(k) < 8Q/5 < L_2(k) < 2Q$, so the parameter interval has length $2Q/35 + O(1)$.

For a prime ℓ , let $\nu_Q(\ell)$ be the number of residues $k \pmod{\ell}$ for which $L_1(k)L_2(k) \equiv 0 \pmod{\ell}$.

Lemma 1 (local root law). *For $(Q, 21) = 1$,*

$$\nu_Q(\ell) = \begin{cases} 1, & \ell \in \{2, 3, 7\} \text{ or } \ell \mid Q, \\ 2, & \text{otherwise.} \end{cases}$$

In particular, the pair of forms is admissible.

Proof. Modulo two, both forms in (3) equal $k + a$. Modulo three, $L_1 = a$ is nonzero and L_2 has one root. Modulo seven, L_2 is constant and nonzero: three times that constant is $2Q \pmod{7}$; meanwhile L_1 has one root. Outside $\{2, 3, 7\}$, both forms have one root, and the two roots coincide exactly when the determinant (4) vanishes modulo ℓ . $\square$

Consequently the singular series has the exact shape

$$\mathfrak{S}_{3,7}(Q) = C_{3,7} \prod_{\substack{\ell \mid Q \\ \ell \geq 5}} \frac{\ell - 1}{\ell - 2}, \quad (5)$$

where $C_{3,7} > 0$ contains the fixed factors and the convergent generic Euler product.

3 The sieve density theorem

Theorem 2. *Uniformly for $Q \geq 8$,*

$$E_{3,7}(Q) \ll \mathfrak{S}_{3,7}(Q) \frac{Q}{(\log Q)^2} \ll \frac{Q \log \log(3Q)}{(\log Q)^2}.$$

Consequently $E_{3,7}(Q)/|\mathcal{P}_Q| \rightarrow 0$.

Proof. If $(Q, 21) > 1$, the primitive edge set is empty. Otherwise, for squarefree d the Chinese remainder theorem and interval counting give

$$\#\{k \in I_Q : d \mid L_1(k)L_2(k)\} = \frac{|I_Q|\nu_Q(d)}{d} + O(\nu_Q(d)). \quad (7)$$

Apply the standard Selberg upper-bound sieve of dimension two to (7), choosing a fixed small power of Q as sieve level. The remainder is the standard $O(2^{\omega(d)})$ interval remainder, and Lemma 1 gives (5). This proves the first estimate.

For each variable exceptional prime,

$$\frac{\ell - 1}{\ell - 2} = \frac{\ell}{\ell - 1} \left(1 + \frac{1}{\ell(\ell - 2)} \right).$$

The product of the second factors converges absolutely, while the first product is at most $Q/\varphi(Q) \ll \log \log(3Q)$. This proves the second estimate. Finally, the prime number theorem gives $|\mathcal{P}_Q| = \pi(2Q) - \pi(Q) \sim Q/\log Q$. $\square$

The theorem is one-sided. It neither predicts how often a resonance occurs nor supplies signed cancellation; it only proves that its maximum possible shell density tends to zero.

4 Finite families and the matched-mass consequence

The determinant argument is not special to coefficients 3 and 7.

Theorem 3 (fixed finite resonance families). *Fix a finite set $\mathcal{R}$ of triples (a, b, c) with $a, b > 0$, $(a, b) = 1$, and $c \neq 0$. The total number of prime-shell solutions to*

$$ap + br = cQ, \quad p, r \in \mathcal{P}_Q,$$

over $(a, b, c) \in \mathcal{R}$ is

$$O_{\mathcal{R}}\left(\frac{Q \log \log(3Q)}{(\log Q)^2}\right) = o(|\mathcal{P}_Q|).$$

Hence the union of all incident prime rows has density $o(1)$.

Proof. For one triple, either there is no integral solution or all solutions have the form $p = p_0 + bk$, $r = r_0 - ak$. The determinant of these forms is $br_0 + ap_0 = cQ$. Away from the fixed primes dividing the coefficients, the two local roots coalesce only at primes dividing cQ . The proof of Theorem 2 therefore applies with a triple-dependent constant. A local inadmissibility yields no large prime pair. Sum the bounds over the fixed set $\mathcal{R}$; at most twice as many vertices as edges are incident. $\square$

For completeness, the support statement has a deterministic energy transfer. Suppose the collision energy is

$$E_G = D + 2\Re \sum_{\{p,r\} \in E(G_Q)} c_{p,r} \langle u_p, u_r \rangle,$$

where $|c_{p,r}| \leq C$, the graph degree is at most Δ , and positive row masses have ratio at most κ . If M_{inc} is the mass on incident rows, then Cauchy–Schwarz gives

$$\frac{(D - E_G)_+}{D} \leq C\Delta \frac{M_{\text{inc}}}{D} \leq 2C\Delta\kappa \frac{|E(G_Q)|}{|\mathcal{P}_Q|}. \quad (6)$$

Indeed, $2|\langle u_p, u_r \rangle| \leq \|u_p\|^2 + \|u_r\|^2$, and every incident mass is charged at most Δ times. Each fixed linear equation has degree at most two, so Theorem 3 makes (6) tend to zero for a fixed finite family.

Let $D(Q)$ denote total literal aligned row mass and $M(Q)$ the mass on the first 3–7 matching. The previous sharp mass theorem gives

$$\frac{M(Q)}{D(Q)} \leq 8 \frac{E_{3,7}(Q)}{|\mathcal{P}_Q|}.$$

Corollary 4 (fixed-saving obstruction). *In the literal aligned first-resonance model, $M(Q)/D(Q) \rightarrow 0$. Even if every edge is perfectly anti-aligned, the mechanism cannot produce any fixed positive proportional saving at all sufficiently large scales. In particular, strict $1/400$ is impossible on this scoped branch.*

Theorem 3 also shows why merely adding a fixed number of similar resonance channels cannot repair the density deficit under comparable row masses. It does not address a number of channels growing with Q .

5 Finite certification and adversarial controls

The executable certificate checks the determinant and local root law directly at multiple scales and primes. At $Q = 25$, the determinant is 400, the exceptional prime 5 has one root rather than two, the singular correction is exactly $4/3$, and the unique edge is $(37, 47)$.

A separate implementation scans every $8 \leq Q \leq 32768$. It reproduces 568,308 edges among 52,199,509 prime rows. Table 1 records exact window statistics. They are diagnostics, not evidence for the asymptotic theorem.

Q window	edges	rows	aggregate E/P
8–31	3	115	3/115
32–127	32	1,472	1/46
128–511	338	19,171	338/19171
512–2047	3,681	255,321	409/28369
2048–8191	42,938	3,482,706	21469/1741353
8192–32768	521,316	48,440,724	43443/4036727

Table 1: Exact finite reproduction scan.

The independent checker uses parameter-index enumeration rather than the producer’s prime-list enumeration. Adversarial tests perturb the determinant by one, replace an exceptional one-root prime by a generic two-root prime, and corrupt the Euler factor; all three mutations are detected. Normal and optimized Python modes are required to produce identical output.

6 Conclusion and firewall

The first primitive resonance now has a complete progression from exact collision geometry to an asymptotic arithmetic obstruction: its graph is a matching, its saving capacity is at most its matched mass, and a Selberg sieve proves that this mass support has density zero in the literal aligned comparable-row model. The same support obstruction holds for every fixed finite family of linear resonances.

This result is deliberately scoped. It proves no twin-prime lower bound, no signed arithmetic cancellation, no identification of literal aligned rows with actual V59 source masses, and no statement about resonance depth growing with Q . Thus the first-resonance and fixed-finite-family branches are closed, while full Gate B and its strict $1/400$ endpoint remain open globally.

References

- [1] Ben Green. Additive combinatorics. Course notes, Section 2.3. Accessed 2026-08-24.
- [2] Heini Halberstam and Hans-Egon Richert. *Sieve Methods*. Academic Press, 1974.
- [3] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, 2004.
- [4] Zeev Rudnick. Selberg’s sieve—twin primes. Course notes, 2015.